

Hybrid deep learning models for automated fetal health prediction using cardiotocography signals

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ABSTRACT: Fetal health monitoring is essential for ensuring the well-being of both the fetus and the mother during pregnancy. Cardiotocography (CTG), which records fetal heart rate and uterine contractions, is a widely adopted diagnostic tool in prenatal care. This study presents an automated and intelligent fetal health classification framework using deep learning architectures to predict fetal status as Normal, Suspect, or Pathological, based on CTG-derived features. With the proposed methodology researchers use various data pre-processing operations starting with feature normalization and continuing with class balancing through SMOTE followed by feature selection conducted using ANOVA F-score. Several deep learning methods are tested during implementation for evaluation. Hybrid MLP+FCNN proved to be the most effective model by reaching 95.47% test accuracy with excellent class-wise precision and recall which demonstrated strong generalization ability. The Hybrid CNN+LSTM model managed to detect pathological patterns from the brain images while achieving an F1-score of 0.94. The study pushes AI-assisted prenatal diagnostic development and provides fundamental methods for operating real-time fetal screening technologies in both medical facilities and remote areas.

Keywords: Cardiotocography (CTG), Deep Learning, Fetal Health, Feature Selection, Automated Diagnosis

1 INTRODUCTION

Success in childbirth depends on proper fetus health maintenance during pregnancy which also minimizes possible perinatal complications [1]. Medical practitioners can enhance diagnosis precision and prenatal care management through early identification of fetal developmental abnormalities [2]. Artificial Intelligence together with Deep Learning technology has developed new possibilities to process CTG data with automated analysis that attains high accuracy in real-time. The deep learning model learns intricate non-linear patterns directly from complex high-dimensional medical datasets without requiring human intervention to engineer features [3,4]. Numerous studies have proven through their positive performance results that deep learning applications succeed in healthcare tasks including disease detection and image understanding and physiological signal examination [5,6]. The research implements deep learning structures to estimate fetal wellness through the extraction of features from CTG measurement output. Three conceptual health categories exist for fetal status analysis which use Normal and Suspect and Pathological annotations from the original dataset. Statistical and frequency-domain features extracted from FHR and UC signals make up this dataset that serves well for developing resilient classification models. The methodology applies data preprocessing and SMOTE class balancing together with ANOVA F-score feature selection followed by deep learning model training using MLP, FCNN, CNN, RNN, LSTM, Autoencoder-based systems and CNN+LSTM hybrid

structures. The evaluation of these models operates through accuracy metrics as well as log loss while using ROC-AUC metrics and confusion matrix and classification report. The primary objective of this study is to develop an automated and reliable fetal health classification system that can assist obstetricians and healthcare professionals in early detection of fetal distress, thereby enabling timely interventions and reducing perinatal risks. The proposed system utilizes various deep learning paradigms in order to address the shortcomings of manual CTG interpretation and traditional machine learning models. This research will generate outcomes that can assist in building smart clinical decision support systems (CDSS) for obstetrics within facilities which lack sufficient expert fetal monitoring specialists.

1.1 Need of the study

The clinical utilization of Cardiotocography (CTG) for fetal monitoring faces interpretation challenges because of arbitrary analyses and inconsistent interpretations between observers while standard decision support systems are absent in various healthcare facilities. The diagnostic outcomes become inconsistent when medical experts manually evaluate CTG data especially for doubtful cases within the “Suspect” range. Proper identification of fetal stress during pregnancy remains vital as it prevents multiple severe birth complications including birth asphyxia and premature birth and neonatal death. Rural health facilities alongside other resource-constrained medical settings face challenges because they do not have qualified staff to properly evaluate and analyze CTG monitoring data. AI technology provides the opportunity for automated fetal health prediction yet the priority lies in creating systems which are both accurate and explainable from digitized CTG records. Thus, this study is motivated by the urgent demand for intelligent clinical decision support tools that can enhance obstetric care, minimize risks, and ensure fetal well-being through real-time prediction of fetal health status using deep learning models.

2 LITERATURE REVIEW

Recent advancements in deep learning have significantly improved the reliability and automation of fetal health monitoring systems. Alasmawi *et al.* [1] developed FUSC (Fetal Ultrasound Semantic Clustering) as a new self-supervised deep learning framework that performs semantic clustering of second-trimester ultrasound images for improved fetal anatomy examination. Semantic representation learning through their approach shows capability to minimize large labeled dataset needs for fetal imaging analysis. The extraction of fetal heartbeat locations from abdominal phonocardiograms was successfully achieved through a deep attention-based transformer model which Almadani *et al.* [2] proposed. The model displays outstanding performance when processing temporal relationships along with traditional signal processing methods because of its ability to handle noisy biomedical signals. The researchers of Basak *et al.* [3] produced a 1D-CycleGAN-based model designed for the extraction of fetal ECG signals from maternal ECG signals while preserving the morphological characteristics. The model demonstrates its capability to maintain essential cardiac waveforms while accomplishing signal separation of mixed signals to provide accurate in-time fetal cardiac assessment. For fetal ultrasound plane recognition Gao *et al.* [4] introduced an ensemble architecture with graph enhancement which combines multi-scale structure perception features. The method uses graph learning alongside deep ensemble networks to extract robust spatial features which enhance diagnosis accuracy when applied across different fetal anatomical planes. The authors Gopikrishna *et al.* [5] compared deep learning segmentation models during a study which assessed their ability to detect and measure fetal ventriculomegaly from MRI scans. The research output proves that model architectural design plays a central role when detecting minor fetal deformities with accurate clinical precision. The rising requirement for explainable AI approaches in medical practice led Harikumar *et al.* [6] to research explainable methods for fetal ultrasound plane classification. The research demonstrates the necessity of clear decision-making in clinical deep learning applications since it establishes

trust and usability between medical staff. Hurtado *et al.* [7] developed an ensemble segmentation method for accurate fetal four-chamber view detection in ultrasound tests which showed better segmentation results through multiple model integration. Deep learning technologies enable researchers McCoy *et al.* [8] to shift their method from fetal image analysis to intrapartum fetal heart rate monitoring preventing neonatal acidemia at birth. The researchers demonstrated that the combination of continuous fetal ECG data with recurrent architectures effectively predicts risks during labor which results in a non-invasive risk assessment tool for adverse outcomes prevention.

3 METHODOLOGY

This methodology outlines the systematic pipeline followed to build an effective deep learning-based predictive system for fetal health classification. The designed process addresses class imbalance problems while enhancing feature relevance while establishing generalization capabilities for untested cases.

3.1 Data preprocessing and feature selections

The dataset used in this study is a publicly available Fetal Health dataset derived from Cardiotocography (CTG) signals. It includes 21 numerical features extracted from Fetal Heart Rate (FHR) and Uterine Contraction (UC) signals.

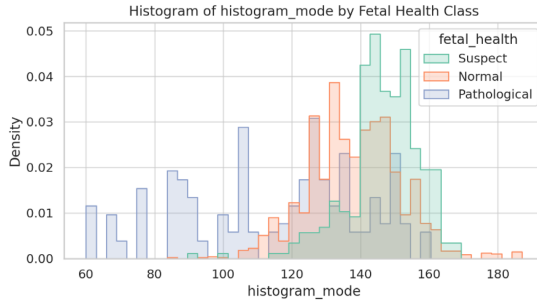


Figure 1. Most frequent fetal health value histogram.

Figure 1 displays the density plots of the `histogram_mode` feature for three fetal health classes. `histogram_mode` likely represents the most frequent fetal heart rate value within a CTG record. The Normal class has peaks around 140–150, with most values falling in the 130–160 range. The Suspect class is also concentrated around 135–150, but with a slightly broader spread compared to the Normal class. In contrast, the Pathological class is left-skewed, covering a much wider and lower range from approximately 60 to 150, with notable density below 120. Thus, the `histogram_mode` feature provides good class separation, especially between the Normal and Pathological cases. The significant left shift for Pathological records suggests that this feature could be highly discriminative in identifying compromised fetal health. A target class labeled as Normal, Suspect and Pathological. This multiclass classification setup enables automated prediction of fetal well-being, which is crucial for obstetric monitoring. All features are standardized using Z-score normalization to ensure that each feature contributes equally as per the given eqn. (1).

$$x_{i_scaled} = (x_i - \mu_i) / \sigma_i \quad (1)$$

The purpose of feature scaling is to avoid bias towards with larger scales. The dataset suffers from an imbalanced class distribution. To address this, SMOTE (Synthetic

Minority Over-sampling Technique) is applied, which generates synthetic examples for minority classes using given eqn. (2).

$$x_{new} = x + \delta * (x_{nearest} - x), \delta \sim U(0, 1) \quad (2)$$

Further, the SelectKBest method with ANOVA F-score (as mentioned in the eqn. 3) is employed to select the top 10 most relevant features as shown in the Figure 2.

$$F = \left[\sum n_j (\bar{x}_j - \bar{x})^2 \right] / \left[\sum \sum (x_{ij} - \bar{x}_j)^2 \right] \quad (3)$$

Where $\bar{x}_j$ is mean of feature in class j and $\bar{x}$ is overall feature mean

This reduces dimensionality, training time, and overfitting risk.

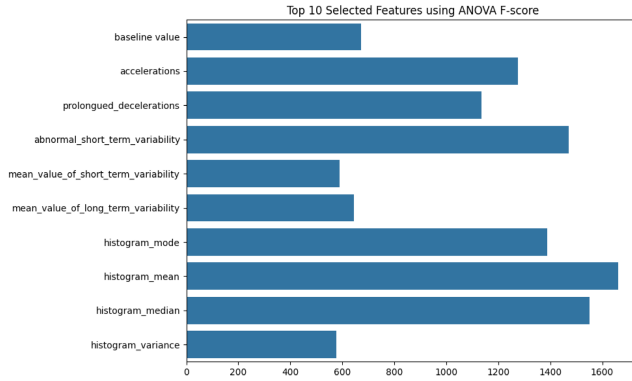


Figure 2. Top selected features using Anova F-score.

3.2 Deep learning model training and development

To classify fetal health status from CTG-derived features, several deep learning models were developed and evaluated. Each model brings unique strengths in capturing spatial, temporal, or combined feature representations. Each model is trained for 50 epochs with a batch size of 32, and a validation split of 20% is used during training. The model optimizes weights through back-propagation using the Adam optimizer. A simple deep learning architecture made up of layers where each neuron is connected to every neuron in the next layer. MLP learns patterns based on the full feature space. No spatial or temporal assumptions are made. It performs well when features are already engineered or not spatially/temporally related as mentioned in the eqn. 4.

$$\begin{aligned} z^{(l)} &= W^{(l)} * a^{(l-1)} + b^{(l)} \\ a^{(l)} &= f(z^{(l)}) \end{aligned} \quad (4)$$

Whereas, applies 1D convolution filters across sequences (given by eqn. 5). It detects local patterns such as trends, spikes, or drops in CTG features over time. CTG signals have a sequential structure (time-based). CNN captures short-term dependencies and repeated local patterns (e.g., accelerations or decelerations in heart rate).

$$\begin{aligned} y_i &= f\left(\sum w_j * x_{\{i+j\}} + b\right) \\ \text{Pooling(optional)} : \\ y_{i-pooled} &= \max(x_i, x_{\{i+1\}}, \dots, x_{\{i+p-1\}}) \end{aligned} \quad (5)$$

In an autoencoder, the encoder is used to reduce input dimensionality, and the decoder reconstructs the input during training. The classifier predicts the class from the encoded representation. The model learns a compressed representation (as mentioned in the eqn. 6) of the input, which is then used to train a classifier for final predictions [8].

$$\begin{aligned}
\text{Encoder} : z &= f(W_e * x + b_e) \\
\text{Decoder} : x &\sim = f(W_d * z + b_d) \\
\text{Classifier} : y &= \text{softmax}(W_c * z + b_c)
\end{aligned} \tag{6}$$

Whereas, Hybrid CNN + LSTM, Combines CNN for spatial features and LSTM for temporal dependencies. This model first uses CNN to extract local features and then feeds the output into an LSTM to capture temporal dependencies. CNN extract local features from sequential input as shown in the below equation.

$$\text{CNN Layer} : y_{\text{CNN}} = \text{Conv1D}(x)$$

Whereas, LSTM process feature sequence over time as given by below equation

$$\text{LSTM Layer} : h_t = \text{LSTM}(y_{\text{CNN}_t})$$

Dense layer + Softmax for output of the hybrid model shown below (given by eqn. 7):

$$\text{Output Layer} : y_{\text{hat}} = \text{softmax}(W * h + b) \tag{7}$$

Hybrid model captures both short-term patterns (with CNN) and long-term dependencies (with LSTM). And it is Well-suited for CTG signals which exhibit both local fluctuations and sequential trends. Among all models, the Hybrid both hybrid models showed superior capability in understanding the nature of CTG signals for fetal health classification on both spatial and temporal features effectively [9,10].

4 RESULTS & DISCUSSION

The performance analysis of different deep learning models contains an evaluation of their metrics across multiple dataset parameters. Analysis results show that the Hybrid MLP+FCNN model delivered the best test accuracy as a sign of its robust capability to use CTG-derived features for fetal health prediction.

Table 1. Training and test results.

Model	Train Accuracy	Train Loss	Val Accuracy	Val Loss	Test Accuracy	Test Loss
FCNN	0.943	0.160	0.943	0.144	0.948	0.155
1D CNN	0.897	0.265	0.933	0.229	0.924	0.218
RNN	0.909	0.235	0.924	0.191	0.926	0.204
LSTM	0.931	0.176	0.934	0.191	0.924	0.238
Autoencoder	0.871	0.336	0.883	0.299	0.895	0.285
Hybrid MLP+FCNN	0.943	0.153	0.953	0.131	0.954	0.154
Hybrid CNN+LSTM	0.905	0.253	0.933	0.198	0.918	0.222

Table 1 shows multiple deep learning model performance analysis for training and validation sets as well as testing. The Hybrid MLP+FCNN model showcased the best test

accuracy measure of 95.47% accompanied by a low-test loss value of 0.1547 that demonstrated strong performance throughout fetal health prediction using CTG-derived features. The model’s capability to match training data points is demonstrated through training accuracy while validation accuracy shows its generalization abilities and test accuracy represents its actual world performance.

Table 2. Classification report summary.

Model	Class	Precision	Recall	F1-Score
FCNN	Normal	0.99	0.94	0.96
	Suspect	0.89	0.97	0.93
	Pathological	0.97	0.94	0.95
1D CNN	Normal	0.98	0.91	0.95
	Suspect	0.84	0.96	0.90
	Pathological	0.97	0.90	0.93
RNN	Normal	0.99	0.92	0.95
	Suspect	0.87	0.92	0.90
	Pathological	0.93	0.94	0.93
LSTM	Normal	0.99	0.91	0.94
	Suspect	0.85	0.96	0.90
	Pathological	0.96	0.91	0.93
Autoencoder	Normal	0.98	0.88	0.93
	Suspect	0.81	0.93	0.86
	Pathological	0.92	0.88	0.90
Hybrid MLP+FCNN	Normal	0.99	0.95	0.97
	Suspect	0.92	0.96	0.94
	Pathological	0.96	0.95	0.95
Hybrid CNN+LSTM	Normal	0.97	0.90	0.94
	Suspect	0.88	0.89	0.88
	Pathological	0.91	0.96	0.94

The models’ performance metrics according to classes appear in Table 2 to provide detailed evaluation results. The evaluation examines three class labels which are Normal, Suspect and Pathological. Among all classification groups the Hybrid CNN+LSTM model displays superior performance especially with a Pathological class F1-score of 0.94. The 1D CNN model displays a challenge predicting the Suspect class because it reaches a precision of 0.84 while other results remain higher. Table 2 functions as a diagnostic tool to determine model performance on individual classes which supports misclassification analysis for erratic classes.

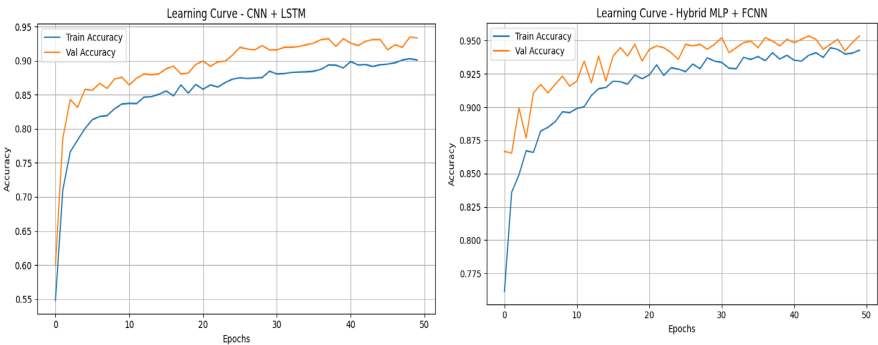


Figure 3. Learning curves for the hybrid deep learning model.

The Figure 3 shows how Hybrid models performed in accuracy measurements for training and validation across 50 epochs. The learning curves provide essential information to evaluate what extent the model learns and how it performs in generalization. The Hybrid MLP + FCNN model demonstrates its suitability for fetal health classification along with its ability to deliver high training and validation accuracy through the learning curve analysis. The model proves suitable for deployment in medical decision systems thanks to its dependable nature.

5 CONCLUSION

In this study, a deep learning-based framework was developed to classify fetal health status into three categories—Normal, Suspect, and Pathological using features derived from cardiotocographic (CTG) recordings. The proposed methodology contained vital preprocessing methods consisting of data normalization together with SMOTE-based class balancing followed by ANOVA F-score feature selection aiming to enhance model reliability. A comprehensive analysis involved testing various deep learning models which included Multilayer Perceptron (MLP) as well as Fully Connected Neural Networks (FCNN) and Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) and Long Short-Term Memory (LSTM) and Autoencoder-based classifiers in addition to more advanced hybrid architectures (such as CNN + LSTM). The hybrid models outperformed others by achieving maximum accuracy and generalization ability through their ability to merge spatial and temporal learning for strong fetal health assessments. Studies revealed that deep learning algorithms successfully retrieved important patterns from CTG datasets with numerous dimensions which allowed them to precisely diagnose fetal health conditions despite imbalanced data. The achieved results create a positive path for developing smart clinical decision support systems (CDSS) which will benefit obstetric care.

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